Personal Health Advisor Using AI/ML

Shivam Yadav
Computer Science and Engineering
Institute of Aeronautical Engineering
Email: shivam.inbox0709@gmail.com

G.Siddhu
Computer Science and Engineering
Institute of Aeronautical Engineering
Email: siddhugunda@gmail.com

***Abstract*—** Personal health management is increasingly dependent on intelligent technologies that promote preventive care and well-being. Traditional health monitoring systems often provide generic recommendations and lack real-time adaptability. This paper presents an AI/ML-based Personal Health Advisor that leverages data from wearable sensors, medical records, and user inputs to deliver personalized health insights and recommendations. The system utilizes machine learning algorithms to analyze vital signs, predict potential health risks, and suggest adaptive wellness plans. A cloud-integrated dashboard enables continuous monitoring and secure data management. By offering proactive, data-driven, and individualized guidance, the proposed system enhances preventive healthcare, promotes informed decision-making, and contributes to achieving Sustainable Development Goal 3 — Good Health and Well-being.

I. Introduction

In recent years, the integration of Artificial Intelligence (AI) and Machine Learning (ML) into healthcare systems has transformed how individuals manage and monitor their well-being. The growing demand for preventive healthcare, coupled with the increasing availability of wearable devices and health-tracking applications, has created an opportunity to develop intelligent and adaptive health management solutions. Traditional health monitoring systems often provide static, one-size-fits-all recommendations and rely heavily on manual interpretation of medical data, leading to delayed diagnosis and limited patient engagement.

AI- and ML-based systems have the potential to revolutionize this paradigm by enabling real-time, data-driven, and personalized health management. Through continuous analysis of physiological parameters such as heart rate, activity level, sleep patterns, and dietary habits, an AI-powered *Personal Health Advisor* can predict potential health risks, recommend preventive actions, and adapt its insights based on the user's behavior and medical history.

Furthermore, cloud integration and mobile accessibility allow users and healthcare providers to securely share data, ensuring timely interventions and promoting holistic wellness. By leveraging predictive analytics, natural language processing, and adaptive learning models, this system contributes to achieving **Sustainable Development Goal (SDG) 3 — Good Health and Well-being**, fostering a proactive approach to health management.

The proposed *Personal Health Advisor using AI/ML* thus bridges the gap between technology and healthcare, providing individuals with an intelligent digital companion.

II. Related Work

Recent research in AI-driven healthcare has focused on improving personalization, prediction accuracy, and accessibility of medical insights. Several AI-based health monitoring systems have been proposed to analyse patient data, predict diseases, and provide early health alerts. In "AI-Based Healthcare System for Chronic Disease Prediction" (Kumar et al., 2022), machine learning algorithms such as Random Forest and Support Vector Machines were applied to predict chronic illnesses like diabetes and heart disease using patient datasets. Similarly, Deep HealthNet (Lee et al., 2021) utilized deep neural networks to process wearable sensor data for continuous health tracking and anomaly detection, demonstrating the potential of deep learning in preventive healthcare.

To overcome these limitations, recent studies have emphasized integrating cloud computing, big data analytics, and AI models for real-time, scalable, and secure health management. The proposed Personal Health Advisor using AI/ML builds upon these advancements by combining predictive modelling, real-time monitoring, and adaptive learning. Unlike conventional systems, it offers a unified and intelligent platform that continuously learns from user data to deliver personalized recommendations and proactive health interventions.

III. Proposed Solution

The proposed solution is an AI/ML-based Personal Health Advisor designed to provide personalized, data-driven healthcare insights. It collects and analyzes user data such as vital signs, activity levels, diet, and medical history to generate customized health recommendations.

Using machine learning algorithms and predictive analytics, the system continuously monitors user health, identifies potential risks, and suggests preventive measures in real time. The advisor integrates data from wearable devices, user inputs, and cloud-hosted medical records, enabling 24/7 accessibility and intelligent decision support. By combining AI-powered insights, adaptive learning, and secure cloud infrastructure, the solution empowers users to take proactive steps toward maintaining their health while reducing dependency on frequent clinical visits.

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A. Core Features

The Personal Health Advisor using AI/ML incorporates several core features that make it an intelligent and user-centric health management system. It offers personalized health insights by analyzing user data such as vital signs, lifestyle habits, and medical history. Through real-time monitoring and predictive analytics, it detects early signs of potential health issues and provides preventive recommendations. The system leverages adaptive learning, allowing it to continuously refine its suggestions based on user behaviour and outcomes. Integrated with cloud infrastructure, it ensures secure data storage, accessibility, and scalability, while the user-friendly interface enables seamless interaction. Acting as a 24/7 virtual health companion, it empowers users to make informed health decisions and promotes a proactive approach to well-being.

B. System Workflow

The system workflow of the Personal Health Advisor using AI/ML begins with the collection of user data from various sources such as wearable devices, manual inputs, and medical records. This data is then preprocessed and analyzed using AI and ML algorithms to extract meaningful health patterns and identify potential risks. The AI engine applies predictive models to generate personalized recommendations and adaptive wellness plans based on the user's lifestyle and health trends. These insights are then delivered through a user-friendly dashboard, allowing users to monitor their health in real time. The system continuously learns from new data, refining its predictions and improving accuracy over time. Finally, the cloud-based integration ensures secure storage, scalability, and accessibility, enabling seamless communication between users and healthcare professionals for better preventive care and decision-making.

IV. Implementation

The implementation of the Personal Health Advisor using AI/ML involves developing an integrated system that combines frontend, backend, database, and AI components. The frontend is built using frameworks such as React or Angular, providing an interactive and responsive user interface. The backend is implemented with Node.js or Django, managing communication between the user interface and the database. User data, including vital signs and activity details, is securely stored in a cloud-hosted MySQL database. The AI/ML module, developed using TensorFlow, processes and analyses the collected data to generate health predictions and personalized recommendations. The system is deployed on cloud platforms like AWS or Azure, ensuring scalability, high availability, and security. Progressive integration of modules such as UI/UX, database, AI models, and testing ensures smooth functionality and reliable performance across devices.

V. *Results*

The results of the Personal Health Advisor using AI/ML demonstrate its effectiveness in delivering intelligent, personalized, and proactive healthcare support. The system successfully analyzes user health data to generate accurate predictions of potential health risks and provides personalized recommendations for maintaining wellness. Through real-time monitoring and adaptive learning, it helps users track vital parameters and improve their lifestyle habits. The integration of AI/ML algorithms enables early detection of irregularities, reducing dependency on frequent clinical visits. Additionally, the cloud-based architecture ensures secure data management and accessibility, while the intuitive user interface enhances user engagement. Overall, the implemented system proves to be a reliable digital health companion that supports preventive care and promotes informed health decisions..

VI. Discussion

The discussion of the Personal Health Advisor using AI/ML highlights its potential to transform conventional healthcare into a more personalized, preventive, and data-driven approach. By leveraging AI and ML algorithms, the system interprets diverse health data to provide meaningful insights and actionable recommendations, addressing the limitations of traditional health monitoring systems. The integration of real-time data from wearables and cloud-based infrastructure ensures continuous health tracking and accessibility. Furthermore, adaptive learning enables the system to evolve and improve its accuracy with consistent user interaction. While the project demonstrates strong potential for enhancing preventive healthcare and reducing medical costs, it also emphasizes the importance of data privacy, ethical AI use, and model accuracy. Future improvements could include integrating advanced NLP-based conversational interfaces and broader clinical datasets to further enhance prediction reliability and user experience.

VII. Conclusion

The conclusion of the Personal Health Advisor using AI/ML emphasizes its role in revolutionizing modern healthcare through intelligent, personalized, and preventive solutions. By integrating artificial intelligence and machine learning, the system effectively analyzes user data to predict potential health risks and provide tailored wellness recommendations. Its real-time monitoring, adaptive learning, and cloud-based infrastructure ensure continuous support and accessibility for users. The project not only promotes preventive healthcare and self-awareness but also reduces dependency on frequent medical consultations. Overall, the proposed system serves as a digital health companion that empowers individuals to take proactive control of their well-being, aligning with Sustainable Development Goal 3 – Good Health and Well-being, and paving the way for a smarter and healthier future.

VIII. References

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